

Advancing Wildlife Image Classification: A Comparative Study of Supervised and Self-Supervised Learning in Tiputini, Ecuador

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Abstract—The loss of biodiversity poses a major global challenge that requires advanced tools for effective wildlife monitoring. Camera traps generate large datasets, but the high cost and time required for manual annotation reduces their utility. This paper investigates the application of supervised and self-supervised learning approaches to the classification of wildlife images using data from the Tiputini Biodiversity Station in Ecuador, a hotspot of global biodiversity. We evaluate multiple supervised models and compare them against two self-supervised frameworks (DINOv2 and SimCLR), while facing challenges such as limited labeled data, imbalanced classes, environmental variability, and poor image quality. A systematic comparison of results shows that a Vision Transformer (ViT-base) model, fine-tuned on the full labeled dataset, achieves the highest classification accuracy of 90.96%. On the other hand, DINOv2 (90.54%) and SimCLR (89.93%) show comparable, although slightly lower, performance in this full-data scenario but excel when trained on smaller subsamples of labeled data, but excel when trained on smaller subsamples of labeled data. DINOv2 surpasses supervised ViT by +6.5pp, +1.9pp, and +2.5pp at 10%, 25%, and 50% labeled data subsamples, respectively (all statistically significant). This demonstrates SSL’s robustness under heavy label scarcity – achieving 89.1% accuracy (95% CI [88.17, 90.14]) with only 10% labels. Our work enables reallocating labeling effort to new species/tasks within a reasonable accuracy sacrifice for the target species. Key contributions include exhaustive hyperparameter optimization, rigorous SSL-vs-supervised comparison, and computational trade-off analysis. The full implementation of this work is available at our GitHub repository¹.

Index Terms—Foundational Models, Supervised Learning, Self-Supervised Learning, Vision Transformers (ViT), Wildlife Image Classification.

I. INTRODUCTION

BIODIVERSITY loss is an increasing global concern given the current rate of species extinction, estimated to be tens of times higher than the historical average over the past ten million years [1]. Effective biodiversity monitoring is essential for assessing human impact on habitats and species [2]. Camera traps have emerged as a widely used tool for non-invasive wildlife monitoring,

generating vast amounts of visual data from biodiversity hotspots like the Tiputini Biodiversity Station (TBS) in Ecuador [3], [4].

However, the sheer volume of collected data, combined with the high cost and time required for manual annotation, represents a significant challenge. Deep learning models have shown promising results in automating wildlife image classification, offering scalable solutions to these challenges [5], [6], [7]. Although the final goal is to monitor population trends and species dynamics, much of the current research remains focused on improving classification accuracy [8].

The limited availability of high-quality labeled datasets has constrained progress, particularly in biodiversity-rich regions like TBS, where existing models have struggled to achieve satisfactory results [4]. This study aims to address these limitations by leveraging transfer learning with state-of-the-art architectures and exploring self-supervised learning techniques to improve classification performance on the TBS dataset.

A. Context and Problem Statement

Biodiversity monitoring is the cornerstone to understand ecosystems’ health, to assess the impact of environmental changes, and inform conservation strategies, especially in the face of climate change and wide animal species extinction [2], [9]. Advances in camera trap technology have revolutionized wildlife monitoring, enabling large-scale, non-invasive data collection in remote ecosystems [3], [10]. However, the scale of these datasets poses significant challenges, particularly, given that data annotation is costly, time-consuming, and prone to error [7], [11], [2].

Deep learning has emerged as a powerful framework for automating image classification reducing the dependence on manual annotation [12], [10]. Supervised learning approaches, particularly, Convolutional Neural Networks (CNNs) like ResNet [13] and EfficientNet [14], have demonstrated strong performance in general image classification tasks. Vision Transformers (ViT), which are based on attention mechanisms, have shown superior performance to CNNs in capturing fine-grained features for specific tasks [15].

Despite their success, applying these models to wildlife classification remains challenging due to three primary factors:

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¹https://github.com/DiegoDVillacreses/ssl_image_classification_tiputini/

- **Data scarcity:** Supervised learning models require extensive labeled datasets, which are often unavailable in biodiversity monitoring due to the high cost of annotation. This reliance on annotated data represents a bottleneck in scaling Deep Learning based solutions for real-world biodiversity monitoring tasks [11], [7]. [12] states that addressing the scarcity of labeled data requires developing methods to leverage large amounts of unlabeled data such as unsupervised or self-supervised learning.
- **Domain generalization:** Models trained on datasets such as ImageNet often fail to generalize to ecological data due to domain shifts in species and environmental contexts, limiting their applicability across diverse ecosystems [10], [11].
- **Environmental factors:** Wildlife images are frequently low resolution, poorly lit, and occluded, further complicating classification tasks and requiring models robust to noisy inputs [3], [4].

B. Research Context: Tiputini Biodiversity Station

The Tiputini Biodiversity Station (TBS), located in Yasuní National Park, Ecuador, is a globally recognized hotspot of biodiversity, with hundreds of species recorded in a single square kilometer [16]. Over 97,000 images have been captured through camera traps at TBS, providing a rich dataset for ecological research. However, only 5,214 images have been labeled, focusing on two primary species: *Taypec* (65.1%) and *Taytaj* (34.9%). The classification of species in Tiputini presents unique challenges [4]:

- **Visual similarity:** The two species exhibit subtle morphological differences, making it difficult to differentiate them even for human annotators (Image 1).
- **Imbalanced data:** The dataset is skewed, with fewer images of *Taytaj* compared to *Taypec*, which can bias model training and evaluation. To mitigate this, undersampling of the majority class was employed, though alternative techniques such as oversampling or weighted loss functions could be explored in future work.
- **Environmental variability:** Images often include occlusions, poor focus, and diverse lighting conditions, reflecting the complexities of field data collection.
- **Non-independence of data:** Each image may contain multiple animals, which introduces the risk of having related instances (e.g., two animals from the same image) appearing in both training and testing sets during cross-validation. This could lead to data leakage. To address this, all train-validation splits were performed using photo IDs as the partitioning key ensuring that all animals from the same image are kept exclusively within either training, validation or testing set.

This paper begins by reviewing prior work in wildlife image classification, then we describe our experimental approach,

which covers the dataset, preprocessing steps, and model configurations. After that we present an analysis of the results, comparing the performance of supervised and self-supervised learning frameworks.

II. STATE OF THE ART

A. Deep Learning

Supervised Deep Learning methods remain the predominant paradigm in computer vision, excelling in image classification when sufficient labeled data is available [17], such as Imagenet [18]. When labeled data is limited, alternative approaches tend to show better results, a careful literature review highlights four alternative approaches: i) transfer learning, ii) semi-supervised learning, iii) self-supervised learning and iv) active learning [19], [20], [21], [22]. A brief discussion about these approaches is included:

- **Transfer learning** uses models pretrained on large datasets and fine-tunes them for specific tasks with limited labeled data [23]. This approach reduces the need for extensive domain-specific data, making it particularly effective in scenarios like wildlife monitoring, where annotated datasets are small. Its advantages include reduced training time and improved performance on small datasets (compared against pure supervised learning). However, transfer learning assumes that the pretrained features are transferable to the target domain, which may not always hold true, particularly when the source and target domains differ significantly [24]. Recent research shows that the excessive usage of ImageNet is leading to overfitting and poor generalization when test images are *harder* than train images [25].
- **Semi-Supervised learning** leverages a small amount of labeled data from a large unlabeled dataset [26]. Two common approaches are: i) *Self-Training*: trains a model on labeled data, pseudo-labels unlabeled samples, and iteratively retrains with the expanded dataset; ii) *Label Propagation*: models data as a graph, spreading labels to unlabeled nodes based on similarity. However, this group of models heavily depends on the assumption that the unlabeled data follows a similar distribution as the labeled data [27].
- **Self-Supervised Learning** uses a pretext tasks to learn from unlabeled data, enabling the model to acquire meaningful representations independently of human labeling [28], [29]. Frameworks such as SimCLR have shown good results on benchmark datasets with a simple implementation, narrowing the performance gap with supervised methods [28]. DINOv2, a state of the art model, shows a remarkable improvement (89.6% against 76.5% top-1 accuracy) over SimCLR but at the cost of code and computational complexity [29]. The main advantage of self-supervised learning is its ability to exploit large-scale unlabeled datasets, which are more easily available. However, the selection and refinement of pretext tasks and its computational requirements can be significant challenges.

- **Active Learning** aims to minimize labeling effort by identifying the most informative samples for annotation. By iteratively asking a human annotator to label the most uncertain samples, active learning reduces the overall data labeling cost [30]. This framework has shown promising results in domains like medical diagnostics [31] or biodiversity monitoring [20]. However, the iterative nature of the process can be restrictive when the labeling team is no longer available.

Considering that we no longer have access to a team that could correctly label more images Active Learning is not a viable option. Additionally, in this study, the primary distinction between Supervised Learning and Transfer Learning lies in the percentage of frozen layers. Therefore, throughout this document, the term "Supervised Learning" includes both approaches unless otherwise specified.

B. Challenges of Camera-Trap Data

Based on the methodologies reviewed in the previous subsection, we now proceed to discuss the applications of deep learning in wildlife classification. While supervised and self-supervised learning offer powerful frameworks for image classification, their deployment in real-world contexts like wildlife monitoring must address the challenges presented by camera-trap data. These difficulties impact data quality and a tailored literature review must be performed to select relevant supervised architectures.

- **Environmental Variability:** Illumination issues, such as poor lighting at night, and weather conditions like fog or rain, often result in low-quality images [3], [5], [32]. Temporal changes, including seasonal variations, exacerbate inconsistencies within the dataset [3].
- **Motion Blur** Fast-moving animals often appear blurred, and some animals occupy only a small fraction of the frame, making feature extraction and classification more difficult [3], [5].
- **Occlusions and Perspective Issues:** Vegetation, environmental obstructions, or other animals frequently occlude the subject. Additionally, different proximity to the camera also increase the difficulty of computer vision tasks [3], [32].
- **False Triggers and Dataset Noise:** Non-animal triggers caused by wind, vegetation, or human activity often dominate datasets, requiring extensive preprocessing to remove irrelevant images [3], [5].
- **Dataset Imbalance:** Wildlife datasets are typically imbalanced, with rare species underrepresented and non-animal images disproportionately frequent [5], [32].
- **Camera Malfunctions:** Equipment issues, such as lens discoloration or hardware failures, introduce additional artifacts and noise into the dataset [3].

To address these challenges, specialized preprocessing techniques or human-driven image selection are often

required. Also, a rigorous approach to result analysis is required to ensure reliable results.

1) *Supervised Learning Applications in Wildlife Classification:* Supervised Learning is the dominant paradigm for computer vision tasks, including wildlife classification, thanks to its ability to extract meaningful features from image data [12]. In the context of wildlife images obtained from camera traps, Supervised Learning address the complex challenges mentioned earlier when paired with Data Augmentation techniques. The use of Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) has become prevalent, showing promising performance across diverse datasets.

Convolutional Neural Networks (CNNs). CNNs have been extensively employed in wildlife classification, the following architectures have shown promising results:

- **ResNet:** [11] reported an 83% accuracy using ResNet-50 (with a similar result using VGG-16) on a dataset compiled from camera traps in Gorongosa National Park, Mozambique. This dataset contains over 111 thousand images from 20 species and did not suffer from occlusion or heavy blurring, although it faced some challenges related to lighting conditions. Additionally, [10] demonstrated that ResNet-101 achieved an accuracy of 91.5% on the Snapshot Serengeti dataset for an image classification task. This dataset consists of 3.2 million camera trap images, including 48 labeled species. However, the data suffered from issues such as heavy imbalanced classes, blurring due to animal movement, varying lighting conditions, and differences in the distances of the targets.
- **VGG:** [32] leveraged VGG-16 with transfer learning to achieve an accuracy of 89.12%. The dataset comprised over 33 thousand labeled images from 19 species captured via camera traps in the Ladakh region of India. The authors do not discuss the complete extent of image-related challenges but mention that, as in other studies, issues as changing weather, improper illumination, and obstructed camera angles were present. They addressed the most problematic images by manually deleting outliers. Similarly, [22] employed the Missouri Camera Traps dataset, which consists of approximately 20,000 images distributed across six animal classes. Class imbalance and environmental variability, including differences in lighting, weather conditions, and backgrounds, made the classification task challenging. The authors found that VGG-16 achieved the best classification accuracy at 69.5%. As previously mentioned, [11] observed nearly identical results when comparing ResNet-50 and VGG-16.
- **EfficientNet:** Although EfficientNet is less frequently used than other architectures, it demonstrates strengths in training speed and achieves superior results on certain datasets. [5] applied EfficientNet to rank 7th in the iWildCam 2019 competition. The corresponding dataset contains over 243,000 training

images representing 14 animal species. Key challenges included class imbalance, poor image quality, and regional variations.

Vision Transformers (ViTs): The introduction of Vision Transformers (ViTs) has marked a important shift in wildlife classification, moving from traditional CNNs to Transformer based architectures. Recent research demonstrating their superior performance on certain datasets. For example, [33] highlights the application of ViTs in marine animal classification, achieving higher accuracy (90%) than ResNet-50 on a dataset of marine wildlife containing 600 training samples across eight species. The dataset exhibits significant class imbalance, with images often displaying blurring and cluttered backgrounds. Similarly, [34] utilized the OpenAnimalTracks dataset, a challenging classification dataset designed to predict animal species based solely on their tracks. This dataset comprises 3579 images from 18 different species and suffers from severe class imbalance. ViT outperformed ResNet-50, VGG-16, and EfficientNet, achieving an accuracy of 68.01%.

C. Self-Supervised Learning

Although Supervised Learning is the dominant paradigm for computer vision tasks, when facing label scarcity Self-Supervised Learning could help to significantly improve accuracy [12], [28]. A recent literature review of 350 papers concluded that self-supervised learning plays a critical role in the development of GPT models, given its ability to effectively leverage unlabeled data [35]. For camera trap images, [36] shows an improvement using SimCLR over Supervised Learning in the Serengeti dataset, the same used by [10], reaching an accuracy of 94.4%, almost 3 percentage points improvement compared against ResNet-101. Other studies, like [37] evaluate various Self-Supervised Learning methods applied to camera trap datasets. The authors compare three methods: SimCLR, MoCo v2, and a novel temporal positive selection (TPS) method. On the Snapshot Serengeti dataset, TPS achieves the best results, with an accuracy of 64.2% using only 1% of labeled data, outperforming SimCLR (61.8%) and MoCo v2 (62.1%). At 10% labeled data, TPS reaches 76.2%, surpassing SimCLR (74.8%) and MoCo v2 (75.4%). This study concludes that Self-Supervised Learning hold great promise for biodiversity monitoring as unlabeled datasets expand.

At the time of writing this document, there is limited DINOv2 finetuning applied to wildlife image classification. Documents like [38] compare ImageNet pretrained DINOv2 against a custom SSL model, finding that DINOv2 performs well in multiple datasets, but, their own model is superior. Other authors like [39] use ImageNet pretrained DINOv2 with a FAISS database to retrieve cropped images based on the embeddings extracted from the SSL model, achieving 1 pp less than their best supervised finetuned model. [40] compare fully finetuned DINOv2 against their own Multi-Crop Attention (MCA) which is a supervised model built upon on DINOv2 that uses attention mechanisms

to understand the context of other animal detections within the same image, in this study MCA shows better performance of 3.34 pp on Serengeti dataset and 2.23 pp on Safari2024.

Based on this literature review, we will fine-tune two self-supervised vision models: DINOv2 and SimCLR. SimCLR will serve as our SSL baseline based on its simpler implementation. Because we are particularly interested in performance on small labeled datasets, we will evaluate both models on progressively smaller subsamples of our annotated data. This will allow us to understand how far SSL can be pushed in order to save labeling effort while maintaining an acceptable drop in classification accuracy.

D. Supervised and Self-Supervised Algorithms

Considering the literature review from the last subsection, we are going to train ResNet, VGG, EfficientNet, Vision Transformers (ViT), SimCLR and DINOv2. This subsection provides a technical overview of those algorithms.

- **ResNet (Residual Networks):** introduced by [13], focus on reducing the vanishing gradient problem by introducing residual connections. These connections facilitate the training of deep architectures by enabling the learning of identity mappings, which allows a stable gradient movement during backpropagation. The backbone of ResNet's architecture consists of: i) a residual block, which bypass the input to an intermediate layer; ii) a convolutional layer; iii) batch normalization; iv) ReLU activation functions. ResNet-152 achieves a top-1 accuracy of **77.0%** on the ImageNet dataset [18].
- **VGG (Visual Geometry Group Networks):** developed by [41], is characterized by its use of small convolutional filters (3x3) throughout the network, combined with max-pooling layers to gradually reduce spatial dimensions and increase the number of feature channels, enabling the extraction of more abstract and complex features at each stage. This uniform configuration enhances feature hierarchy learning and simplifies the design compared to earlier architectures. VGG16 achieves a top-1 accuracy of **71.5%**, while VGG19 reaches **71.6%** on ImageNet.
- **EfficientNet:** proposed by [42], introduces a compound scaling method to systematically scale network dimensions—depth, width, and input resolution—using a fixed scaling coefficient. The architecture is based on mobile inverted bottleneck convolutions (MBConv) and incorporates squeeze-and-excitation blocks for improved channel-wise feature recalibration. The EfficientNet-B7 variant achieves a top-1 accuracy of **84.4%** on ImageNet, while being significantly more computationally efficient than traditional architectures like ResNet and VGG.
- **Vision Transformers (ViT):** Vision Transformers (ViT), introduced [15], uses self-attention instead of convolutional layers. Self-attention consist in dividing

input images into fixed-size subsets, which are linearly represented into a sequence and then processed by a transformer decoder. The self-attention mechanism enables ViT to model to learn dependencies between all subsets within the input. Pretrained on ImageNet-21k and fine-tuned on ImageNet, ViT achieves a top-1 accuracy of **84.0%** with the ViT-B/16 configuration.

- **SimCLR (Simple Framework for Contrastive Learning)**: introduced by [28], is a self-supervised model that learns by maximizing the similitude between augmented frames (heavily distorted views of the original image) of the same image while separating frames of different images in a latent space. SimCLR achieves this objective by optimizing a contrastive loss. When fine-tuned on ImageNet, SimCLR achieves a top-1 accuracy of **76.5%** using a ResNet-50 encoder, showing the ability to rival supervised algorithms in computer vision classification tasks.
- **DINOv2 (Distillation-based Self-Supervised Vision Transformer)**: proposed by [29], is a self-supervised model that trains a student ViT by matching a momentum-updated teacher on differently augmented views of the same image. DINOv2's learned features are highly discriminative: after a linear classifier on ImageNet-1k it gets **86.5%** top-1 accuracy.

E. Agnostic Approach to Training

Given the uncertainty about how pretrained features align with the unique characteristics of biodiversity datasets, this study adopts an agnostic approach to model training:

- 1) **Fully Retraining All Layers**: When labeled data is sufficiently available, retraining models like ResNet, MobileNet, EfficientNet, or ViT from scratch allows them to learn task-specific representations. However, this approach requires significant computational resources and risks overfitting in scenarios with limited data [12].
- 2) **Transfer Learning with Partial Fine-Tuning**: Transfer learning is particularly useful when labeled data is scarce or computational resources are limited to re-train the full architecture. Given the pretrained weights, it keeps knowledge from low-level features, such as edges and textures, while fine-tuning the last layers. When fine-tuning, the model learn high-level features of the data that usually represent data-specific information [17].
- 3) **Self-Supervised Learning with SimCLR**: offers an alternative by leveraging unlabeled data [28]. We are going to use ViT base and ViT large as encoder, retraining the full model while experimenting with different batch sizes and number of neurons per layer on the MLP head.
- 4) **Self-Supervised Learning with DINOv2**: offers an state of the art model that leveraging only unlabeled data [29]. We trained three models with different hyperparameters.

III. MATERIALS AND METHODS

A. Materials

1) *Dataset Description*: The datasets utilized in this study were collected at the Tiputini Biodiversity Station (TBS) in Yasuní National Park, Ecuador, one of the most biodiverse regions in the world [16]. These datasets are categorized into labeled and unlabeled subsets:

Labeled Dataset:: This dataset is the same as the one used on [4], it comprises of 5,214 images manually annotated and selected by biology experts. The dataset includes two species: *Taypec* and *Taytaj*, species selected due to their physical similarity, posing a particularly difficult task for computer vision. These images were captured using camera traps. All labeled images were resized to a fixed resolution of 416x416 pixels, as our objective is image classification we use the provided bounding boxes to crop the images.



Figure 1: Representative images of labeled data

Key characteristics of the labeled dataset:

- **Species Distribution**: The dataset is imbalanced, with 65.1% of images classified as *Taypec* and 34.9% as *Taytaj*.
- **Animal Count per Image**: *Taypec* images contain an average of 2.88 animals, with a maximum of 12, whereas *Taytaj* images feature an average of 1.71 animals, with a maximum of 7.
- **Diversity in Capture Conditions**: Images include a range of lighting conditions, times of day, and varying environmental backgrounds. Characteristics handpicked in order to provide the most diverse and challenging dataset.

Unlabeled Dataset:: The unlabeled dataset is significantly larger, containing approximately 97,000 high-resolution images captured through the same camera traps. Unlike the labeled dataset, it includes images of around 70 different, unidentified species, which remain unlabeled [4]. These images were processed using the MegaDetector-v5, model applied for this particular dataset by [43], which identified 45,000 frames containing wildlife and provided a bounding box, we only use the information within the bounding box. We used an arbitrary cut-point of 85% of confidence in order to select images with animals. This cut-point could be used as a hyperparameter in future studies.

Details about the unlabeled dataset:

- **Resolution Variability:** The images retain their original high-resolution format, which varies across the dataset. This higher resolution aids in detecting fine-grained details but adds computational time to the analysis.
- **Species Diversity:** The dataset features a diverse array of species and environmental conditions, which enhances its value for self-supervised learning approaches like SimCLR, where feature representation from a wide variety of contexts increases the probability of the model to improve accuracy in validation and test sets.
- **Role in Study:** This dataset complements the labeled subset by providing a rich source for unsupervised training.

2) *Hardware and Software Specifications:* The computational experiments were conducted using advanced hardware and software configurations to allow for the dataset size and complexity:

- **Hardware:**
 - NVIDIA H200 GPU with 140 GB of VRAM.
- **Software:**
 - **Frameworks and Libraries:**
 - * Python: 3.10.15
 - * PyTorch: 2.5.1+cu12.1
 - * Torchvision: 0.20.1+cu12.1
 - * PyTorch Lightning: 2.4.0
 - * Transformers: 4.46.3
 - * PIL (Pillow): 11.0.0
 - * Pandas: 2.2.2
 - * Numpy: 1.26.4
 - **Operating System:**
 - * Ubuntu 22.04.4 LTS (GNU/Linux 5.15.0-124-generic x86_64)

3) *Preprocessing Techniques:* To optimize the datasets for training and evaluation, distinct preprocessing pipelines were applied to the labeled and unlabeled datasets, tailored to their respective roles in the study.

Unlabeled Dataset:

For the unlabeled dataset, a preprocessing pipeline was designed to maximize the effectiveness of SimCLR based on the recommendations from [44] a follow up paper from the original where states a possible best preprocessing for SimCLR.

- **Random Horizontal Flip:** Flips the image horizontally with a 50% probability.
- **Random Resized Crop:** Crops a random portion of the image and resizes it to 224×224 pixels, introducing positional variation.
- **Random Color Jitter:** Applies brightness, contrast, saturation, and hue adjustments with a probability of 0.8. Particularly:
 - Brightness, contrast, and saturation: randomly modified within $\pm 50\%$.

- Hue: Randomly modified within ± 0.1 .

- **Random Grayscale:** Introduces grayscale with a probability of 0.2, simulating images with reduced color information.
- **Gaussian Blur:** Introduces a Gaussian blur with a kernel size of 9, which applies a smoothing effect, simulating lower image clarity.
- **Normalization:** Normalizes the image pixel values to have a mean of 0.5 and a standard deviation of 0.5 for each channel, improving model convergence.

Labeled Dataset:

Due to the high degree of visual similarity between the *Taypec* and *Taytaj* classes, we adopted a comprehensive data augmentation pipeline to increase intra-class variance and improve generalization performance. Similar approaches have been suggested in prior research, which highlights the effectiveness of augmentation strategies such as random cropping, flipping, color jittering, and Gaussian blurring in handling visually challenging datasets [45]. The following steps were applied:

- **Training Preprocessing:**
 - **Random Resized Crop:** Crops a random region of the image and resizes it to the target size. The scale of the crop is randomly selected between 80% and 100% of the original size, with an aspect ratio chosen between 0.75 and 1.33.
 - **Random Horizontal and Vertical Flip:** Flips the image horizontally and vertically, each with a probability of 0.5.
 - **Random Color Jitter:** Introduces random variations in brightness, contrast, saturation, and hue, with maximum changes of 0.4, 0.4, 0.4, and 0.2, respectively, applied with a probability of 0.8.
 - **Random Grayscale:** Converts the image to grayscale with a probability of 0.2.
 - **Gaussian Blur:** Applies a Gaussian blur with a kernel size of 5 and a sigma range of 0.1 to 2.0, applied with a probability of 0.5.
 - **Random Affine Transformations:** Applies geometric transformations simulating variations in viewpoint and orientation:
 - * **Rotation:** Randomly rotates the image to a maximum of 15 degrees.
 - * **Translation:** Shifts the image horizontally and vertically to a maximum of 10% of its size.
 - * **Scaling:** Scales the image by a random factor between 0.9 and 1.1.
 - * **Shearing:** Applies a shift in the x-axis while keeping y-axis fixed, with a maximum angle of 10 degrees.
 - **Random Erasing:** Randomly erases a subsection of the image with a probability of 0.5. The region encompasses between 2% to 33% of the image area. This region is replaced with random values.
 - **Normalization:** Pixel values were normalized to a mean of $[0.485, 0.456, 0.406]$ and a standard de-

viation of [0.229, 0.224, 0.225], based on ImageNet parameters.

- **Validation and Test Preprocessing:** Validation and test does not use any data augmentation, only formatting for easier computations:
 - **Resizing:** All images were resized to the target size.
 - **Normalization:** mean of [0.485, 0.456, 0.406] and a standard deviation of [0.229, 0.224, 0.225].

B. Methodology

The methodology for this study is structured into four distinct blocks: preprocessing, supervised training, self-supervised training, and performance comparison. Each block is tailored to address the challenging aspects of this particular dataset, using both labeled and unlabeled images.

1) B.1. Data Preprocessing and Splitting:

Unlabeled Dataset: The unlabeled dataset, include approximately 97,000 images captured through camera traps at the Tiputini Biodiversity Station (TBS), was processed using *MegaDetector-v5*. Images with a detection confidence of 85% or higher were extracted, resulting in a filtered subset of 45,000 images. These images, featuring a wide variety of species and environmental conditions, were cropped into the suggested bounding box and then resized to 224×224 pixels, for compatibility with ViT. Data augmentation, as described in the Preprocessing Techniques subsection, was applied for SimCLR, to comply with contrastive learning data requirements.

Labeled Dataset: Two frameworks were used, the first one employ the full labeled dataset, the second one, employ subsamples with the following proportions: [0.1, 0.25, 0.5, 0.6, 0.7, 0.8, 0.9]. For both frameworks the suggested bounding box was used, then, the dataset was downsampled to ensure balanced classes. For the first framework, the balanced dataset was then divided into: i) training set: 70% of the images, ii) validation set: 20% of the images, iii) testing set: 10% of the images. For the second framework, we start joining training and validation sets from the full labeled dataset, then 25 random subsamples were taken for each proportion size, after that, for each subsample a 80-20 train-validation split is performed. For each step special care was taken to avoid data contamination, as many images contain multiple animals.

SSL Task Splitting: For the SimCLR self-supervised contrastive task, the unlabeled dataset was split into: i) training set: 80% of the images, ii) validation set: 20% of the images. DINOv2 does not require a train-test splitting. The labeled dataset was retained exclusively for supervised tasks guaranteeing no data contamination between labeled and unlabeled information.

2) B.2. Supervised Model Training:

We use five select architectures based on the state-of-the-art review: i) ResNet-50, ii) VGG-16, iii) EfficientNet-B7, iv) ViT-Base, v) ViT-Large. Then, hyperparameter optimization in the full labeled dataset was performed using the following hyperparameters:

- **Learning rate:** {0.001, 0.0001}.
- **Batch size:** {32, 64}.
- **Frozen layers:** {0%, 15%, 50%, 80%, 95%}, allowing exploration of transfer learning with total and partial fine-tuning.

Models were trained for up to 500 epochs, with early stopping applied after 30 epochs of no improvement in validation binary cross-entropy loss. Subsamples use best model and hyperparameters, with the exception of Frozen layers. These configurations were applied to 5 subsamples for each subsample size. Since 15% Frozen Layer showed the best mean validation accuracy for all subsample sizes, only for that configuration, we compute the results for the 25 subsamples, for this results, confidence intervals at 95% are computed using a t-distribution adjusted to the sample size.

3) B.3. Self-Supervised Model Training:

Model: SimCLR

- **Hyperparameter Optimization:** Hyperparameters were optimized via grid search:
 - **Encoders:** ViT-base, ViT-Large.
 - **Batch Size:** 64, 128, 256, 512.
 - **Head Size:** 8192, 16384, 32768.
- **Data Augmentation:** as described in the Preprocessing Techniques subsection.
- **Contrastive Loss Optimization:** SimCLR’s contrastive loss was optimized with the following static parameters:
 - **Temperature parameter:** 0.07.
 - **Learning rate:** 5e-5.

Training was conducted for 500 epochs and model patience of 30 epochs.

Model: DINOv2

- Three DINOv2 models were trained. The first model uses the suggested configurations of the original DINOv2 [46] but we changed batch size from 64 to 16 given vRAM restrictions and extend training to 600 epochs.
- For the second model, we try to improve iBOT Loss assigning it a dedicated head. We change the following configurations expecting a smoother training log: i) teacher start momentum was raised from 0.992 to 0.994, ii) decrease the base learning rate from 0.004 to 0.0005, iii) Learning Rate from 0.0005 to $6.25e - 05$.
- For the third model, following the objective of improving iBOT Loss, we kept a dedicated head for that loss

but reverted the rest of the configurations to isolate the effect of the independent training head.

Multilayer Perceptron after SSL: A Multilayer Perceptron -MLP- with two layers, each with the same number of neurons as embeddings from the SSL (1024 for DINOv2 and SimCLR with ViT large and 768 for SimCLR with ViT base). As in the supervised learning part, the Multilayer Perceptron -MLP- has two frameworks: i) for the full sample, ii) for subsamples. For those two parts the MLP was: i) trained on frozen features extracted from the pretrained encoders of SimCLR or DINOv2, ii) Bayesian Hyperparameter Optimization was applied to maximize validation accuracy. For the full sample performance was evaluated using the same validation set to keep consistency with supervised models. The Bayesian Hyperparameter Optimization has the following search space was:

$$\begin{aligned} \text{batch_size} &\in \{16, 32\}, \\ \text{lr} &\in \{5 \times 10^{-6}, 1 \times 10^{-6}, 5 \times 10^{-5}, \dots, \\ &\quad 1 \times 10^{-1}\}, \\ \text{weight_decay} &\in \{0, 5 \times 10^{-6}, \dots, 1 \times 10^{-1}\}, \\ \text{dropout} &\in \{0, 0.1, 0.15, \dots, 0.95\}. \end{aligned}$$

For the full sample one validation computation was performed, for the subsamples, 5 validation computations were performed for all SSL models. After which we got an idea of the two best SSL configurations, those three models were computed for the 25 validation sets for each subsample size, for this results, confidence intervals at 95% are computed using a t-distribution adjusted to the sample size.

4) B.4. Performance Evaluation:

- **Initial Evaluation:** All models were first evaluated using a single train-validation split, using the accuracy as the only metric given the balanced information we get after undersampling for the full sample. For the subsamples, first 5 validation sets were used for all the SSL and supervised models, once we got an idea of the best models, 25 validation sets were used.
- **Cross-Validation:** The top two models from the initial evaluation were subjected to repeated k -fold cross-validation:
 - **Folds:** 5 folds.
 - **Repetitions:** 5 repetitions.

Folds were sampled from the combined training and validation sets of the original labeled dataset and then undersampled.

- **Embedding Visualization:** To compare the learned feature representations, t-distributed Stochastic Neighbor Embedding (t-SNE) was applied. t-SNE provides a visual representation of the embeddings, helping understand class separability and clustering behavior.
- **Final Testing:** The best-performing model from the cross-validation phase was evaluated on the test set, providing a definitive measure of its generalization performance.

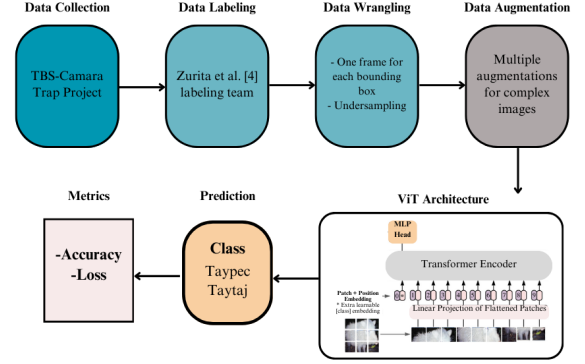


Figure 2: Flow Chart for ViT Training

In order to better illustrate the pipeline of our work we present a graph summarizing a supervised learning example:

IV. RESULTS AND DISCUSSION

A. Quantitative Results

- **Supervised Models:** Within the supervised models, Vision Transformer base (ViT-base) achieved the highest accuracy with **90.96%**. ViT-large and EfficientNet followed with accuracies of **90.85%** and **89.22%**, respectively. ResNet-50 and VGG-16 showed the weakest performance with 85.67% and 87.19% (Figure 5). These results correspond to the best accuracy for each architecture after grid-search, to see all results refer to Table I and II. Since ViT is the best result overall we compute its test accuracy getting a robust value **89.38**.

Training time shows interesting results, ViT-base is the fastest with 10.5 minutes, followed by ResNet with 13.13 minutes, VGG shows similar training time with 15.2 minutes. EfficientNet and ViT-large are the slowest with 21.2 and 34.2 minutes respectively. In this case the best results are also the less computational costly.

Also, ViT-large and ViT-base have almost identical results, but ViT-large is computationally costlier. Additionally, EfficientNet and VGG shows high instability in validation accuracy and higher instability in validation loss (Figure 7).

These results demonstrate the effectiveness of ViT's attention-based architecture in distinguishing between the highly similar classes, despite the limited size of the labeled dataset. Statistical analysis after repeated k -fold cross-validation confirmed that the performance of ViT against EfficientNet is superior ($p < 0.05$).

- **SSL Performance:** For the full labeled dataset, DINOv2 and SimCLR performed near the top supervised ViT results, with an accuracy of **90.65%** and **89.9%** respectively (Table III)². The best DINOv2

²Interested readers can be found in [www.. github](https://www.github.com)

used default configuration, as seen in Figure 9 its iBOT Loss have a very unstable convergence with a gradual decline followed with a stabler but small increase. Considering this erratic behavior we trained a second model with a separate head for the iBOT loss, although it showed a stabler convergence it didn't showed an overall decrease (Figure 9). The second DINOv2 model underperformed (89.0%) the first DINOv2 model and the best SimCLR, we find similar problems with our third DINOv2.

Following the full sample analysis, SimCLR showed an interesting behavior, since we were limited by vRAM we were not able to try big batch sizes as [28], instead, we tried increasing the total number of neurons of each layer in the MLP head. MLP size showed an important effect in some cases, for example for ViT base (batch size of 512) it improves accuracy from 89.43% (16,384 MLP layer size) to 89.9% (32,768 MLP layer size) as seen in Table III. We find that SimCLR with ViT large (its training log is in Figure 8), 256 batch size and 32,768 MLP per layer neurons showed a very near performance to the best DINOv2. This is an interesting result since it shows different conclusions from the standard literature, were DINOv2 outperforms SimCLR with a wide margin.

- SSL Subsample Performance:** To further understand batch size effect on SimCLR please refer to Figure 3, we choose a subsample of 50% since smaller labeled dataset are more informative for SSL. In the graph the blue bars represent ViT base encoder, the red points represents ViT large and the black vertical lines the confidence intervals, we couldn't compute ViT large for 512 batch size with 32,768 MLP layer size for vRAM restrictions, all results were computed for 5 samples. It can be seen that the validation accuracy generally increases with larger batch sizes and head dimensions, peaking at 256 with 32,768 neurons which also shows one of the smallest confidence interval. Also note while ViT Base models (red) show comparable performance to ViT Large at batch size 512 with overlapping (but bigger) confidence intervals.
- SSL against Supervised Performance for sub-samples:** The self-supervised DINOv2 model outperforms the fully supervised ViT-Base on small labeled subsets: after Bayesian hyperparameter optimization of a two-layer MLP, DINOv2 achieves 89.1% vs. 82.7% at 10% sampling, 89.3% vs. 86.8% at 25%, and 89.7% vs. 87.0% at 50%. Beyond the 50% subsample point both models converge around 89–90% with overlapping 95% confidence intervals (Figure 4). For exact values of Confidence Intervals please refer to Table IV
- Frozen Layers Analysis:**

The best validation accuracy for ViT-base is achieved when most of its layers are retrained (15% frozen layers), suggesting that the features learned from ImageNet are not very informative for these camera traps images. Observation which is further supported by the reduction in accuracy across all four supervised

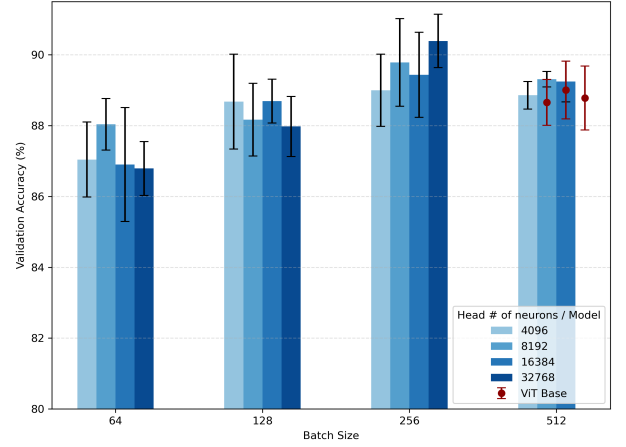


Figure 3: SimCLR Validation Accuracy per Batch Size and Neurons in MLP for a subsample of 50%

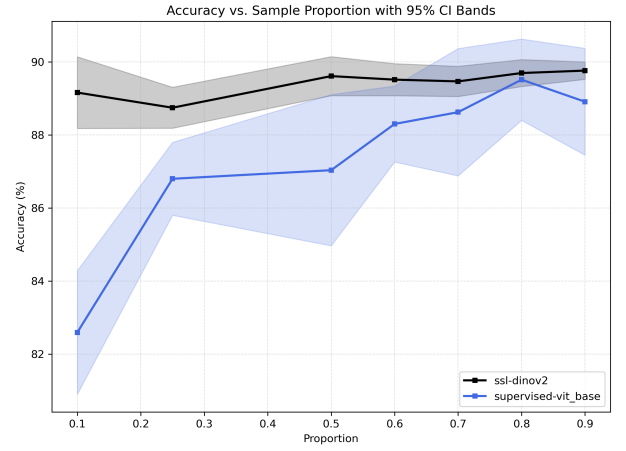


Figure 4: Best SSL against best supervised model per subsample size.

models as more layers are frozen. Also, SimCLR and DINOv2 re-trains the full ViT architecture achieving relative good results.

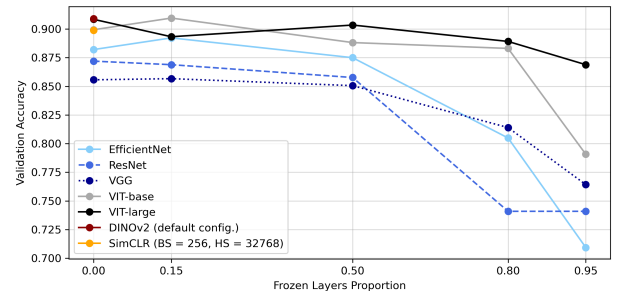


Figure 5: Best Accuracy per architecture and percentage of frozen layer: SimCLR vs. Supervised Learning Models.

B. Qualitative Results

- Embedding Visualization (t-SNE):** The t-SNE visualization of learned embeddings (computed on

validation dataset) shows differences between ViT and SimCLR. ViT’s embeddings form an almost linearly separable space, showing its ability to learn features to differentiate between the two species. ViT’s embeddings exhibit a relatively smooth dispersion of points, suggesting the absence of well-defined clusters beyond the separation of the target classes. In contrast, SimCLR’s embeddings produce a more complex and less separable space, showing limitations when learning a representation that captures the species differences.(Figure 6).

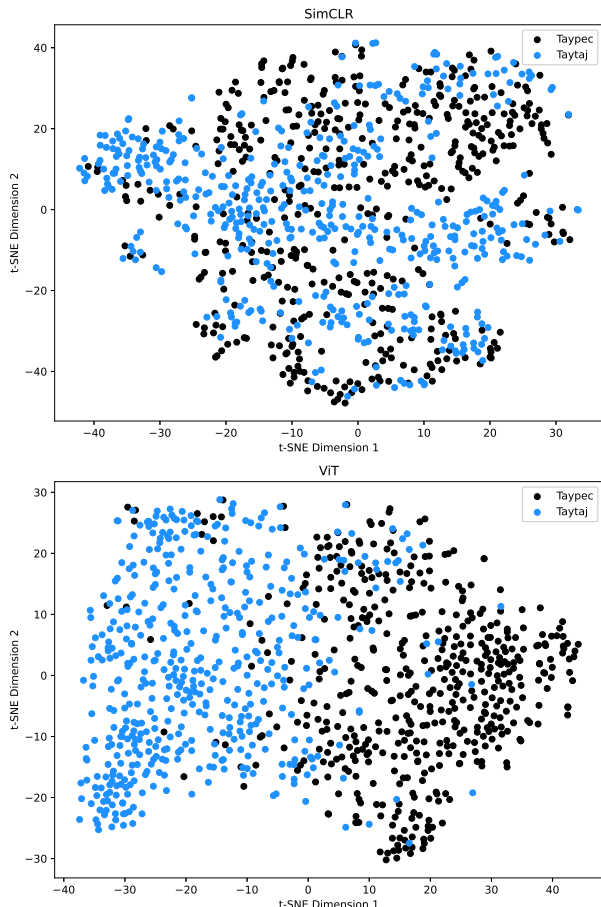


Figure 6: Embedding Visualization (t-SNE). Comparing SimCLR against best Supervised Model (ViT).

- **Error Analysis:** In order to perform an analysis of misclassifications we introduced a k-means clustering to the t-SNE space from ViT’s embeddings (Figure 11). Despite the smooth structure of ViT’s t-SNE, we identified 15 clusters. Insights were found from five randomly selected images per cluster (Figure 12): i) Clusters 0 and 5 are the more distant to the apparent linear separation of classes, those images are clear with no occlusion or blurring; ii) Cluster 2 contains most of the non separable samples, which correspond to very low quality frames; iii) Cluster 6 and 12 are also challenging regions, its images show blurring, occlusion and poor lightning. Despite the

poor quality of the images the classifier only shows 9.9% of error, highlighting again the robustness of ViT in the presence of suboptimal input conditions.

C. Conclusions

The results clearly show the advantages of ViT’s ability to detect small details and separate images based on them within a complex ecological dataset, demonstrating strong performance against CNN despite its reliance on a small labeled dataset. The results from DINOv2 are robust across all subsample sizes used. The performance on small data fractions is of particular relevance: subsamples of 10% and 25% show accuracies 89.1% with a 95% Confidence Interval of [88.17, 90.14] and 88.74% and a CI of [88.18, 89.30] respectively.

The applicability of our results to other biodiversity datasets may face challenges due to domain-specific factors such as differing species characteristics and environmental conditions. Further experimentation is needed to address these challenges.

D. Future Work

Potential directions for future research can be explored to advance this:

- **Self-Supervised Algorithms Exploration:** Investigate and implement alternative self-supervised learning algorithms like MCA from [40].
- **Active Learning for Labeling:** Incorporate active learning strategies to improve labeling efficiency if possible during the labeling process with biology experts.

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REFERENCES

- [1] T. Jeff, “Humans are driving one million species to extinction,” *Nature*, vol. 569, p. 171, 2019.
- [2] F. Urbano, R. Viterbi, L. Pedrotti, E. Vettorazzo, C. Movalli, and L. Corlatti, “Enhancing biodiversity conservation and monitoring in protected areas through efficient data management,” *Environmental Monitoring and Assessment*, vol. 196, no. 1, p. 12, 2024.
- [3] F. Trollet, C. Vermeulen, M.-C. Huynen, and A. Hambuckers, “Use of camera traps for wildlife studies: a review,” *Biotechnologie, Agronomie, Société et Environnement*, vol. 18, no. 3, 2014.
- [4] M.-J. Zurita, D. Riofrío, N. Pérez, D. Romo, D. S. Benítez, R. F. Moyano, F. Grijalva, and M. Baldeon-Calisto, “Towards automatic animal classification in wildlife environments for native species monitoring in the amazon,” in *2023 IEEE Colombian Conference on Applications of Computational Intelligence (Col-CACI)*. IEEE, 2023, pp. 1–6.

- [5] A. Abuduweili, X. Wu, and X. Tao, "Efficient method for categorize animals in the wild," *arXiv preprint arXiv:1907.13037*, 2019.
- [6] B. H. Curtin and S. J. Matthews, "Deep learning for inexpensive image classification of wildlife on the raspberry pi," in *2019 IEEE 10th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*. IEEE, 2019, pp. 0082–0087.
- [7] L. Bothmann, L. Wimmer, O. Charrakh, T. Weber, H. Edelhoff, W. Peters, H. Nguyen, C. Benjamin, and A. Menzel, "Automated wildlife image classification: An active learning tool for ecological applications," *Ecological Informatics*, vol. 77, p. 102231, 2023.
- [8] G. S. Ferrante, F. M. Rodrigues, F. R. Andrade, R. Goularte, and R. I. Meneguette, "Understanding the state of the art in animal detection and classification using computer vision technologies," in *2021 IEEE International Conference on Big Data (Big Data)*. Ieee, 2021, pp. 3056–3065.
- [9] G. Chen, T. X. Han, Z. He, R. Kays, and T. Forrester, "Deep convolutional neural network based species recognition for wild animal monitoring," in *2014 IEEE international conference on image processing (ICIP)*. IEEE, 2014, pp. 858–862.
- [10] M. S. Norouzzadeh, A. Nguyen, M. Kosmala, A. Swanson, M. S. Palmer, C. Packer, and J. Clune, "Automatically identifying, counting, and describing wild animals in camera-trap images with deep learning," *Proceedings of the National Academy of Sciences*, vol. 115, no. 25, pp. E5716–E5725, 2018.
- [11] Z. Miao, K. M. Gaynor, J. Wang, Z. Liu, O. Muellerklein, M. S. Norouzzadeh, A. McInturff, R. C. Bowie, R. Nathan, S. X. Yu *et al.*, "Insights and approaches using deep learning to classify wildlife," *Scientific reports*, vol. 9, no. 1, p. 8137, 2019.
- [12] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [14] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 4510–4520.
- [15] A. Dosovitskiy, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.
- [16] The Swifttest, "Biodiversity index," <https://theswifttest.com/biodiversity-index/>, accessed April 13, 2024.
- [17] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, 2010.
- [18] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 248–255.
- [19] S. V. Mahadevkar, B. Khemani, S. Patil, K. Kotecha, D. R. Vora, A. Abraham, and L. A. Gabralla, "A review on machine learning styles in computer vision—techniques and future directions," *IEEE Access*, vol. 10, pp. 107 293–107 329, 2022.
- [20] M. S. Norouzzadeh, D. Morris, S. Beery, N. Joshi, N. Jojic, and J. Clune, "A deep active learning system for species identification and counting in camera trap images," *Methods in ecology and evolution*, vol. 12, no. 1, pp. 150–161, 2021.
- [21] Z. Zhao, L. Alzubaidi, J. Zhang, Y. Duan, and Y. Gu, "A comparison review of transfer learning and self-supervised learning: Definitions, applications, advantages and limitations," *Expert Systems with Applications*, vol. 242, p. 122807, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957417423033092>
- [22] N. Sheikh, "Identification and classification of wildlife from camera-trap images using machine learning and computer vision," Ph.D. dissertation, Dublin, National College of Ireland, 2020.
- [23] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Transactions on knowledge and data engineering*, vol. 22, no. 10, pp. 1345–1359, 2009.
- [24] K. Weiss, T. M. Khoshgoftaar, and D. Wang, "A survey of transfer learning," *Journal of Big data*, vol. 3, pp. 1–40, 2016.
- [25] B. Recht, R. Roelofs, L. Schmidt, and V. Shankar, "Do imagenet classifiers generalize to imagenet?" in *International conference on machine learning*. PMLR, 2019, pp. 5389–5400.
- [26] J. E. Van Engelen and H. H. Hoos, "A survey on semi-supervised learning," *Machine learning*, vol. 109, no. 2, pp. 373–440, 2020.
- [27] L.-Z. Guo, Z.-Y. Zhang, Y. Jiang, Y.-F. Li, and Z.-H. Zhou, "Safe deep semi-supervised learning for unseen-class unlabeled data," in *International conference on machine learning*. PMLR, 2020, pp. 3897–3906.
- [28] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *International conference on machine learning*. PMLR, 2020, pp. 1597–1607.
- [29] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby *et al.*, "Dinov2: Learning robust visual features without supervision," *arXiv preprint arXiv:2304.07193*, 2023.
- [30] P. Ren, Y. Xiao, X. Chang, P.-Y. Huang, Z. Li, B. B. Gupta, X. Chen, and X. Wang, "A survey of deep active learning," *ACM computing surveys (CSUR)*, vol. 54, no. 9, pp. 1–40, 2021.
- [31] S. Budd, E. C. Robinson, and B. Kainz, "A survey on active learning and human-in-the-loop deep learning for medical image analysis," *Medical image analysis*, vol. 71, p. 102062, 2021.
- [32] S. S. Kathait, A. Kumar, P. Dhuliya, and I. Chauhan, "Deep learning-based model for wildlife species classification," Valiance Analytics Pvt. Ltd. and Doon University, Noida, Uttar Pradesh, Tech. Rep., 2024, accessed: 2024-07-21. [Online]. Available: <https://valiancesolutions.com/wp-content/uploads/2024/02/Deep-Learning-based-Model-for-Wildlife-Species-Classification.pdf>
- [33] Y. Kesani and D. Mahato, "Advancements in marine animal classification through computer vision and machine learning," *ResearchGate*, 2023, accessed: 2024-11-24. [Online]. Available: https://www.researchgate.net/profile/Deepshikha-Mahato/publication/376683947_Advancements_in_Marine_Animal_Classification_through_Computer_Vision_and_Machine_Learning/links/65839da16f6e450f198d14a4/Advancements-in-Marine-Animal-Classification-through-Computer-Vision-and-Machine-Learning.pdf
- [34] R. Shinoda and K. Shiohara, "Openanimaltracks: A dataset for animal track recognition," *arXiv preprint arXiv:2406.09647*, 2024.
- [35] K. S. Kalyan, "A survey of gpt-3 family large language models including chatgpt and gpt-4," *Natural Language Processing Journal*, vol. 6, p. 100048, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2949719123000456>
- [36] A. Tendle, "“the revolution will not be supervised”: An investigation of the efficacy and reasoning process of self-supervised representations," *ResearchGate*, 2021, accessed: 2024-11-24. [Online]. Available: https://www.researchgate.net/publication/351625872_The_Revolution_Will_Not_Be_Supervised_An_Investigation_of_the_Efficacy_and_Reasoning_Process_of_Self-Supervised_Representations
- [37] O. Pantazis, G. J. Brostow, K. E. Jones, and O. Mac Aodha, "Focus on the positives: Self-supervised learning for biodiversity monitoring," in *Proceedings of the IEEE/CVF International conference on computer vision*, 2021, pp. 10 583–10 592.

- [38] V. Čermák, L. Pícek, L. Adam, and K. Papafitsoros, “Wildlife-datasets: An open-source toolkit for animal re-identification,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 5953–5963.
- [39] J. Vyskočil and L. Pícek, “Towards zero-shot camera trap image categorization,” in *European Conference on Computer Vision*. Springer, 2024, pp. 37–53.
- [40] G. Dussert, S. Dray, S. Chamaillé-Jammes, and V. Miele, “Paying attention to other animal detections improves camera trap classification models,” *bioRxiv*, pp. 2025–07, 2025.
- [41] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” *arXiv preprint arXiv:1409.1556*, 2014.
- [42] M. Tan and Q. Le, “Efficientnet: Rethinking model scaling for convolutional neural networks,” in *International conference on machine learning*. PMLR, 2019, pp. 6105–6114.
- [43] O. Cajamarca and F. Grijalva, “Clustering wildlife species in the amazon: Using vision transformers to analyze unlabeled images from the tiputini biodiversity station,” 2024, unpublished Master’s thesis, Universidad San Francisco de Quito - Master’s Degree in Data Science.
- [44] T. Chen, S. Kornblith, K. Swersky, M. Norouzi, and G. E. Hinton, “Big self-supervised models are strong semi-supervised learners,” *Advances in neural information processing systems*, vol. 33, pp. 22 243–22 255, 2020.
- [45] C. Shorten and T. M. Khoshgoftaar, “A survey on image data augmentation for deep learning,” *Journal of Big Data*, vol. 6, no. 60, pp. 1–48, 2019. [Online]. Available: <https://journalofbigdata.springeropen.com/articles/10.1186/s40537-019-0197-0>
- [46] F. A. Research, “`ssl_default_config.yaml`: Dinov2 self-supervised learning default configuration,” https://github.com/facebookresearch/dinov2/blob/main/dinov2/configs/ssl_default_config.yaml, 2023, gitHub repository, commit ID: (accessed 2025-08-04).

APPENDIX A
ADDITIONAL RESULTS

Table I: Grid Search Results for Supervised Architectures (Part 1)

Model	Learning Rate	Batch Size	Frozen Layers Proportion	Validation Loss	Validation Accuracy	Training Time in Minutes
EfficientNet	0.0001	16	0.00	0.3131	0.8496	23.36
EfficientNet	0.0001	16	0.15	0.2954	0.8648	20.86
EfficientNet	0.0001	16	0.50	0.3283	0.8537	13.60
EfficientNet	0.0001	16	0.80	0.4284	0.7947	43.13
EfficientNet	0.0001	16	0.95	0.5898	0.6860	49.25
EfficientNet	0.0001	32	0.00	0.3010	0.8638	15.48
EfficientNet	0.0001	32	0.15	0.2992	0.8618	15.30
EfficientNet	0.0001	32	0.50	0.3162	0.8638	10.48
EfficientNet	0.0001	32	0.80	0.4488	0.7785	25.37
EfficientNet	0.0001	32	0.95	0.5890	0.6850	47.69
EfficientNet	0.0001	64	0.00	0.3028	0.8821	14.82
EfficientNet	0.0001	64	0.15	0.3023	0.8852	15.53
EfficientNet	0.0001	64	0.50	0.3300	0.8537	10.75
EfficientNet	0.0001	64	0.80	0.4261	0.8008	51.33
EfficientNet	0.0001	64	0.95	0.5887	0.7002	42.19
EfficientNet	0.0010	16	0.00	0.3433	0.8476	38.48
EfficientNet	0.0010	16	0.15	0.3314	0.8699	44.52
EfficientNet	0.0010	16	0.50	0.3415	0.8567	20.39
EfficientNet	0.0010	16	0.80	0.4465	0.7917	23.87
EfficientNet	0.0010	16	0.95	0.5774	0.7043	23.70
EfficientNet	0.0010	32	0.00	0.3454	0.8577	25.58
EfficientNet	0.0010	32	0.15	0.3215	0.8648	26.30
EfficientNet	0.0010	32	0.50	0.3284	0.8750	15.92
EfficientNet	0.0010	32	0.80	0.4311	0.8008	25.28
EfficientNet	0.0010	32	0.95	0.5791	0.6972	18.18
EfficientNet	0.0010	64	0.00	0.3181	0.8669	29.48
EfficientNet	0.0010	64	0.15	0.3543	0.8577	17.74
EfficientNet	0.0010	64	0.50	0.3386	0.8608	14.29
EfficientNet	0.0010	64	0.80	0.4226	0.7957	21.92
EfficientNet	0.0010	64	0.95	0.5730	0.7093	21.20
ResNet	0.0001	16	0.00	0.3331	0.8720	13.14
ResNet	0.0001	16	0.15	0.3432	0.8608	11.48
ResNet	0.0001	16	0.50	0.3480	0.8577	7.32
ResNet	0.0001	16	0.80	0.5555	0.7337	28.32
ResNet	0.0001	16	0.95	0.5555	0.7337	27.91
ResNet	0.0001	32	0.00	0.3440	0.8577	8.47
ResNet	0.0001	32	0.15	0.3293	0.8689	8.47
ResNet	0.0001	32	0.50	0.3601	0.8516	7.45
ResNet	0.0001	32	0.80	0.5632	0.7114	19.41
ResNet	0.0001	32	0.95	0.5632	0.7114	19.27
ResNet	0.0001	64	0.00	0.3467	0.8506	6.41
ResNet	0.0001	64	0.15	0.3608	0.8374	7.10
ResNet	0.0001	64	0.50	0.3598	0.8333	6.65
ResNet	0.0001	64	0.80	0.5763	0.7063	21.56
ResNet	0.0001	64	0.95	0.5763	0.7063	19.94
ResNet	0.0010	16	0.00	0.3836	0.8079	36.82
ResNet	0.0010	16	0.15	0.4079	0.8028	29.92
ResNet	0.0010	16	0.50	0.3904	0.8232	18.18
ResNet	0.0010	16	0.80	0.5301	0.7327	14.99
ResNet	0.0010	16	0.95	0.5301	0.7327	14.61
ResNet	0.0010	32	0.00	0.3717	0.8262	22.04
ResNet	0.0010	32	0.15	0.3867	0.8283	23.34
ResNet	0.0010	32	0.50	0.3761	0.8537	22.56
ResNet	0.0010	32	0.80	0.5406	0.7297	7.79
ResNet	0.0010	32	0.95	0.5406	0.7297	6.97
ResNet	0.0010	64	0.00	0.3823	0.8272	11.81
ResNet	0.0010	64	0.15	0.3626	0.8506	22.99
ResNet	0.0010	64	0.50	0.3779	0.8537	17.65
ResNet	0.0010	64	0.80	0.5343	0.7246	13.67
ResNet	0.0010	64	0.95	0.5343	0.7246	13.27

Table II: Grid Search Results for Supervised Architectures (Part 2)

Model	Learning Rate	Batch Size	Frozen Layers Proportion	Validation Loss	Validation Accuracy	Training Time in Minutes
VGG	0.0001	16	0.00	0.3586	0.8394	21.83
VGG	0.0001	16	0.15	0.3665	0.8333	16.86
VGG	0.0001	16	0.50	0.4063	0.8506	11.22
VGG	0.0001	16	0.80	0.4463	0.7937	9.96
VGG	0.0001	16	0.95	0.5228	0.7480	20.88
VGG	0.0001	32	0.00	0.3600	0.8537	15.38
VGG	0.0001	32	0.15	0.3550	0.8567	15.20
VGG	0.0001	32	0.50	0.3747	0.8445	9.15
VGG	0.0001	32	0.80	0.4323	0.7988	11.87
VGG	0.0001	32	0.95	0.5301	0.7368	12.46
VGG	0.0001	64	0.00	0.3533	0.8557	12.81
VGG	0.0001	64	0.15	0.3455	0.8486	12.05
VGG	0.0001	64	0.50	0.3462	0.8486	9.78
VGG	0.0001	64	0.80	0.4319	0.8079	15.13
VGG	0.0001	64	0.95	0.5272	0.7287	21.64
VGG	0.0010	16	0.00	0.6931	0.5000	28.13
VGG	0.0010	16	0.15	0.6931	0.5000	10.18
VGG	0.0010	16	0.50	0.6931	0.5000	18.72
VGG	0.0010	16	0.80	0.4615	0.7886	15.68
VGG	0.0010	16	0.95	0.5064	0.7480	12.73
VGG	0.0010	32	0.00	0.6931	0.5000	11.74
VGG	0.0010	32	0.15	0.6931	0.5000	11.30
VGG	0.0010	32	0.50	0.6931	0.5000	12.58
VGG	0.0010	32	0.80	0.4504	0.8039	12.88
VGG	0.0010	32	0.95	0.4992	0.7642	12.31
VGG	0.0010	64	0.00	0.6931	0.5000	6.91
VGG	0.0010	64	0.15	0.6873	0.5376	10.35
VGG	0.0010	64	0.50	0.6928	0.5091	10.85
VGG	0.0010	64	0.80	0.4545	0.7876	9.22
VGG	0.0010	64	0.95	0.5086	0.7490	8.77
ViT-base	0.0001	16	0.00	0.2769	0.8811	13.38
ViT-base	0.0001	16	0.15	0.2564	0.8974	12.37
ViT-base	0.0001	16	0.50	0.2862	0.8841	9.42
ViT-base	0.0001	16	0.80	0.2948	0.8831	10.40
ViT-base	0.0001	16	0.95	0.4825	0.7754	9.42
ViT-base	0.0001	32	0.00	0.2610	0.8963	9.32
ViT-base	0.0001	32	0.15	0.2411	0.9096	10.51
ViT-base	0.0001	32	0.50	0.2795	0.8852	8.47
ViT-base	0.0001	32	0.80	0.3572	0.8374	8.00
ViT-base	0.0001	32	0.95	0.4918	0.7693	7.49
ViT-base	0.0001	64	0.00	0.2449	0.8994	10.02
ViT-base	0.0001	64	0.15	0.2382	0.9055	9.33
ViT-base	0.0001	64	0.50	0.2703	0.8882	9.87
ViT-base	0.0001	64	0.80	0.3286	0.8557	8.79
ViT-base	0.0001	64	0.95	0.5099	0.7642	7.48
ViT-base	0.0010	16	0.00	0.5692	0.7093	31.07
ViT-base	0.0010	16	0.15	0.5735	0.6992	19.78
ViT-base	0.0010	16	0.50	0.3454	0.8476	10.71
ViT-base	0.0010	16	0.80	0.3356	0.8587	11.49
ViT-base	0.0010	16	0.95	0.4656	0.7856	6.33
ViT-base	0.0010	32	0.00	0.5591	0.7185	29.61
ViT-base	0.0010	32	0.15	0.3291	0.8618	23.91
ViT-base	0.0010	32	0.50	0.3279	0.8557	7.97
ViT-base	0.0010	32	0.80	0.3140	0.8669	8.03
ViT-base	0.0010	32	0.95	0.4705	0.7734	5.67
ViT-base	0.0010	64	0.00	0.5827	0.6809	20.89
ViT-base	0.0010	64	0.15	0.3306	0.8506	20.38
ViT-base	0.0010	64	0.50	0.3177	0.8709	9.54
ViT-base	0.0010	64	0.80	0.3088	0.8720	7.73
ViT-base	0.0010	64	0.95	0.4630	0.7907	6.09
ViT-large	0.0001	16	0.00	0.6236	0.6372	37.71
ViT-large	0.0001	16	0.15	0.2592	0.8923	27.98
ViT-large	0.0001	16	0.50	0.2541	0.9035	24.10
ViT-large	0.0001	16	0.80	0.2739	0.8882	17.49
ViT-large	0.0001	16	0.95	0.2970	0.8598	19.03
ViT-large	0.0001	32	0.00	0.2540	0.9085	34.27
ViT-large	0.0001	32	0.15	0.2501	0.8933	24.89
ViT-large	0.0001	32	0.50	0.2526	0.8974	17.01
ViT-large	0.0001	32	0.80	0.3014	0.8709	13.94
ViT-large	0.0001	32	0.95	0.3670	0.8364	12.22
ViT-large	0.0001	64	0.00	0.2334	0.9004	24.16
ViT-large	0.0001	64	0.15	0.2460	0.8923	22.99
ViT-large	0.0001	64	0.50	0.2566	0.8882	20.15
ViT-large	0.0001	64	0.80	0.2825	0.8770	13.80
ViT-large	0.0001	64	0.95	0.3239	0.8598	13.92
ViT-large	0.0010	16	0.00	0.6739	0.5965	34.33
ViT-large	0.0010	16	0.15	0.3990	0.8201	79.95
ViT-large	0.0010	16	0.50	0.3072	0.8638	28.09
ViT-large	0.0010	16	0.80	0.3154	0.8567	26.80
ViT-large	0.0010	16	0.95	0.3356	0.8557	18.28
ViT-large	0.0010	32	0.00	0.6304	0.6555	67.90
ViT-large	0.0010	32	0.15	0.5024	0.7449	40.47
ViT-large	0.0010	32	0.50	0.3365	0.8659	17.27
ViT-large	0.0010	32	0.80	0.2875	0.8831	22.79
ViT-large	0.0010	32	0.95	0.3589	0.8425	11.50
ViT-large	0.0010	64	0.00	0.6839	0.5854	21.76
ViT-large	0.0010	64	0.15	0.6931	0.5000	18.34
ViT-large	0.0010	64	0.50	0.3122	0.8750	27.57
ViT-large	0.0010	64	0.80	0.3271	0.8628	15.20
ViT-large	0.0010	64	0.95	0.3426	0.8506	11.35

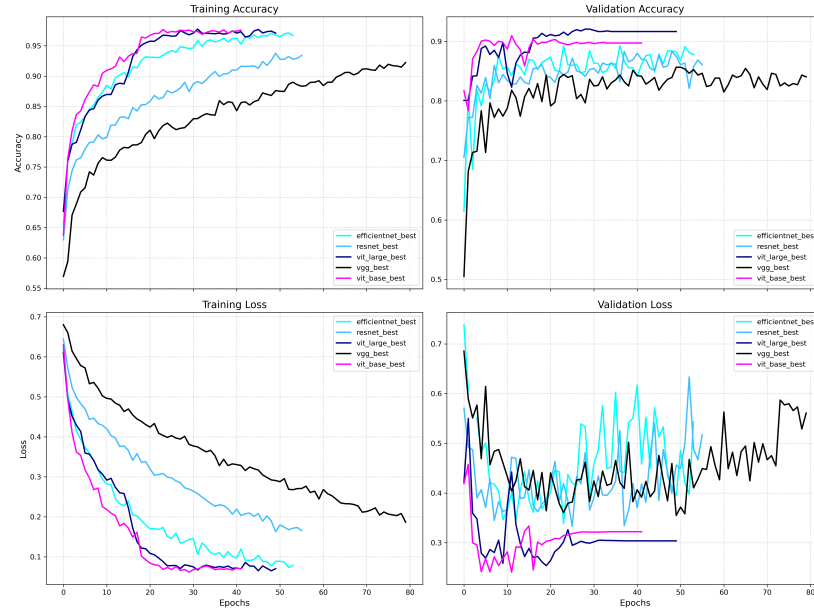


Figure 7: Training Log for best supervised configurations

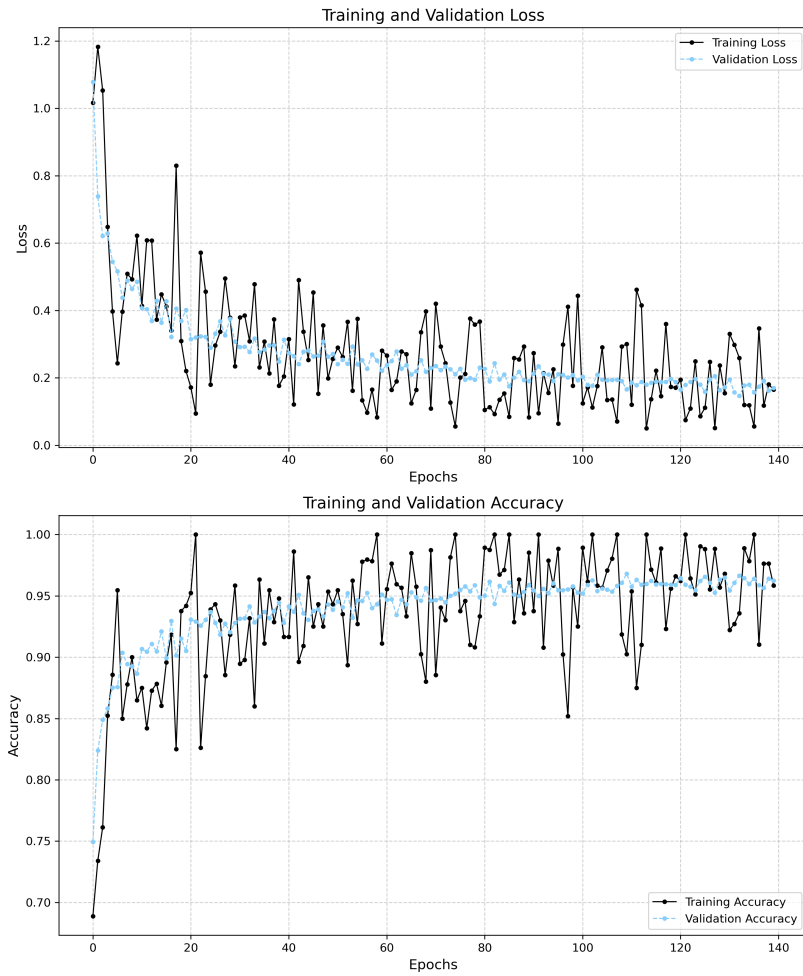


Figure 8: SimCLR Contrastive Training and Validation Results per Epoch.

Figure 9: Training logs for Model 1 of DINOv2

Training Losses Over Iterations

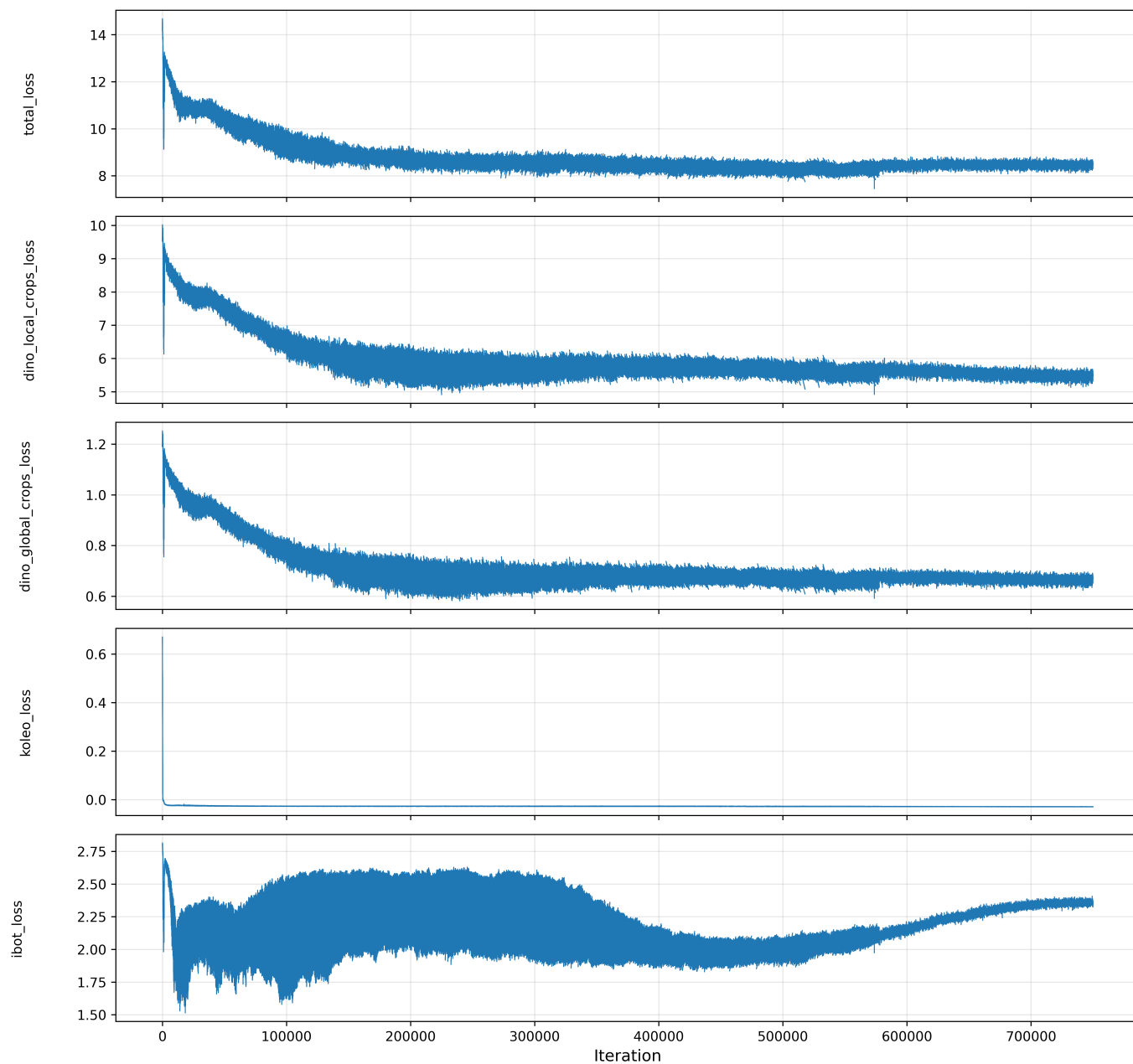


Figure 10: Training logs for Model 2 of DINOv2

Training Losses Over Iterations

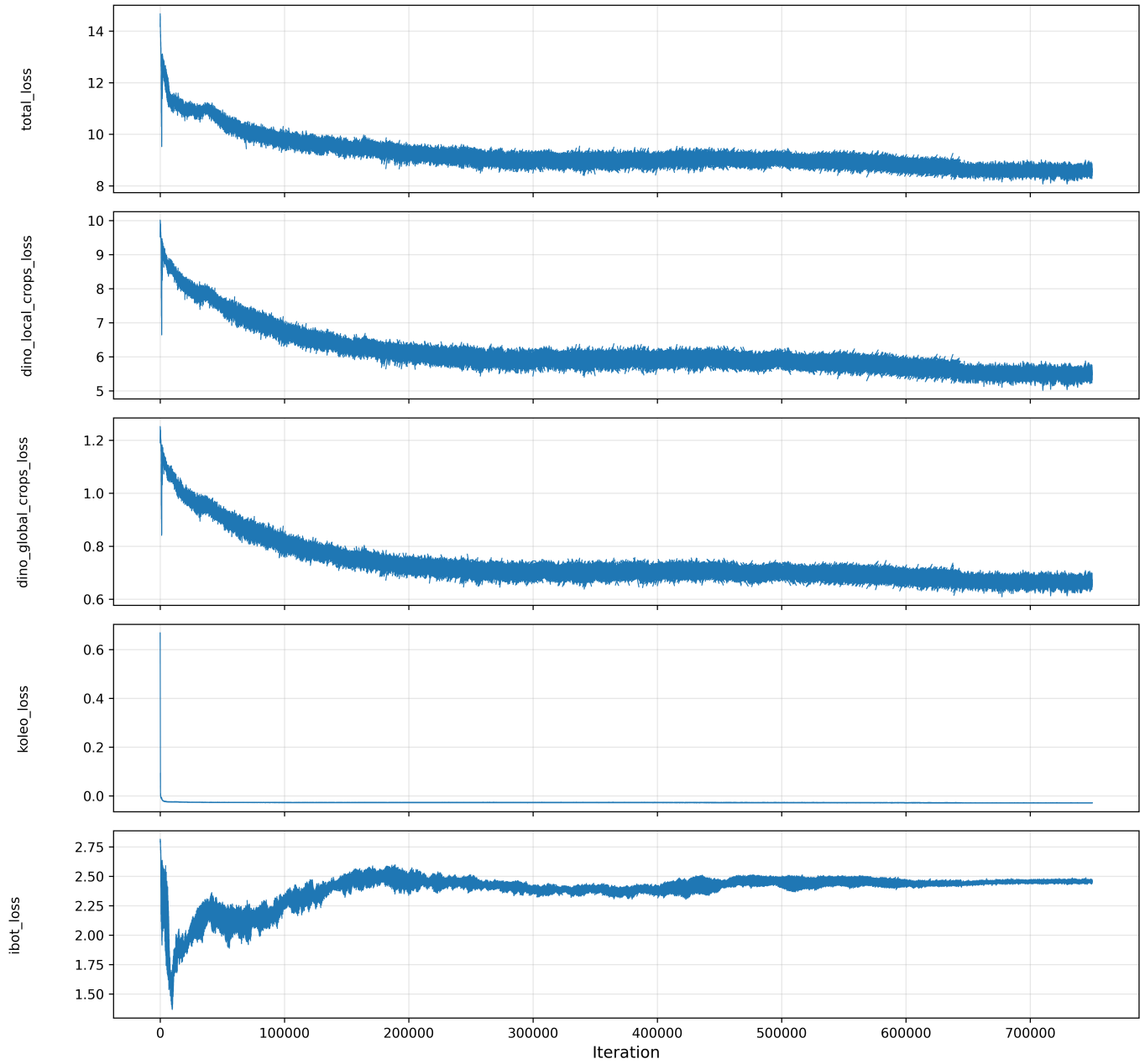


Table III: Model Accuracy per SSL Model

Model	Accuracy	Batch Size	MLP Neurons / Layer
DINOv2 Model1	0.905488	-	-
SimCLR-ViT large	0.899390	256	32768
SimCLR-ViT large	0.897406	256	16384
SimCLR-ViT large	0.896341	256	8192
DINOv2 Model3	0.896341	-	-
SimCLR-ViT base	0.895325	512	16384
SimCLR-ViT large	0.895325	128	32768
SimCLR-ViT large	0.894309	512	8192
SimCLR-ViT base	0.894309	512	32768
SimCLR-ViT base	0.893293	512	8192
SimCLR-ViT large	0.892276	64	8192
DINOv2 Model2	0.891260	-	-
SimCLR-ViT large	0.890244	128	8192
SimCLR-ViT large	0.887195	512	16384
SimCLR-ViT large	0.886179	128	16384
SimCLR-ViT large	0.876016	64	16384
SimCLR-ViT large	0.875000	64	32768

Table IV: Comparison of SSL (DINOv2) and Supervised (ViT Base) Models Across Sample Proportions. Each validation accuracy is reported as the mean across 25 runs with 95% confidence intervals.

Model Type	Model Name	Sample Proportion	Val. Accuracy (%)	95% CI Lower	95% CI Upper	# Runs for CI
SSL	DINOv2	0.10	89.16	88.18	90.14	25
		0.25	88.75	88.18	89.31	25
		0.50	89.61	89.08	90.15	25
		0.60	89.51	89.08	89.95	25
		0.70	89.47	89.05	89.88	25
		0.80	89.70	89.33	90.06	25
		0.90	89.76	89.52	90.00	25
Supervised	ViT Base	0.10	82.59	80.90	84.28	25
		0.25	86.80	85.80	87.80	25
		0.50	87.03	84.96	89.10	25
		0.60	88.30	87.26	89.34	25
		0.70	88.62	86.88	90.37	25
		0.80	89.51	88.40	90.63	25
		0.90	88.91	87.45	90.37	25

Figure 11: Elbow Method Graph and t-SNE Colored by Cluster.

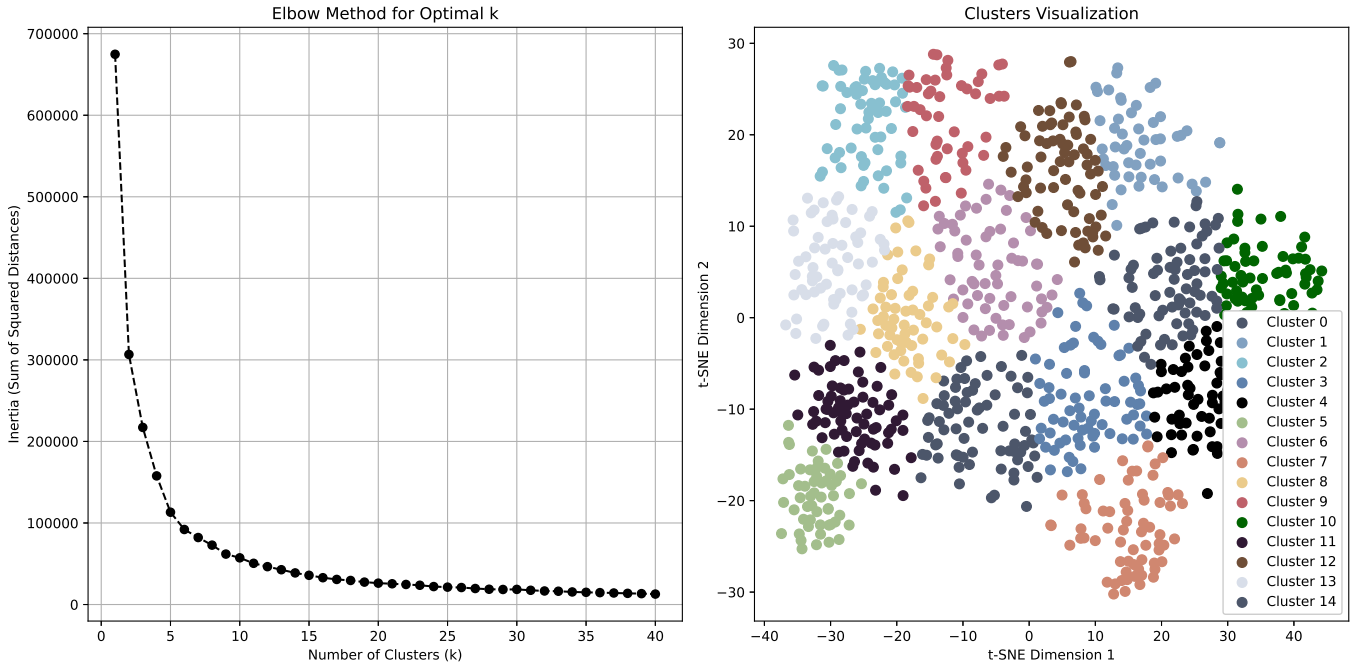


Figure 12: Sample Images from Selected Clusters.

